Javier Raut.

Software Engineer

Cagayan de Oro, Philippines · javier.raut@gmail.com · portfolio.jmraut.dev · github.com/KvassAndVodka · linkedin.com/in/raut-javier-m

SUMMARY

Software engineer who ships full-stack web and map applications to real users: a GIS project-tracking platform for a House of Representatives district office, and my own self-hosted portfolio platform, end to end. BS Computer Science, Magna Cum Laude.

EXPERIENCE

Software Developer

2nd Congressional District Office of Misamis Oriental, House of Representatives · Tagoloan · 2026 – Present

- Took over development of a React 19 + Firebase GIS platform from its original developer that tracks 600+ public works projects; shipped 34 releases and 167 merged pull requests since July 2026.
- Cut about 1.4 s of blocking network work from every page load by profiling production and removing an unused auth redirect resolver; merged duplicate Firestore listeners into one shared subscription.
- Migrated the map from Leaflet to MapLibre GL with self-hosted PMTiles vector tiles, an offline basemap and 3D terrain cached by a service worker, so the app works for field staff on weak signal.
- Rebuilt bulk import as a lock-protected Cloud Functions job queue that processes CSV uploads in bounded, resumable chunks, so concurrent or failed imports cannot corrupt records.
- Acted on a 12-issue security audit: versioned Firestore and Storage rules, a stored-XSS fix, upload validation and role-based access, guarded by 800+ automated tests including an emulator rules suite.
- Built geospatial validation that checks every pin against 504 barangay polygons, and a road-graph editor that snaps project lines to the road network.
- Led a UI/UX overhaul with a shared React Aria component kit and lazy-loaded features.

Part-time Faculty

University of Science and Technology of Southern Philippines · 2026 – Present · Teaching Fundamentals of Database Systems

ML/AI Intern

meldCX PH · Cagayan de Oro · 2025

- Built a working real-time license plate recognition pipeline within a 30-day internship.
- Trained and benchmarked YOLO detection and PaddleOCR models, then optimized inference with OpenVINO and containerized the pipeline with Docker.

PROJECTS

Portfolio and publishing platform

portfolio.jmraut.dev · Next.js 16, React 19, TypeScript, Prisma, PostgreSQL, MapLibre GL, GSAP, Docker

- Scroll-driven 3D map story, a blog, and an admin panel with privacy-preserving analytics (hashed visitor IDs, traffic sources).
- Runs on a home Proxmox server through Docker Compose, a self-hosted GitHub Actions runner and a Cloudflare Tunnel; slimmed Docker images from 2.21 GB to 127 MB.
- Fixed a 1.7 s main-thread freeze on the analytics dashboard by replacing 2,600 per-element animations with one clip animation.

staRT — live transcription service

github.com/KvassAndVodka/staRT · Python, FastAPI, SQLite, FFmpeg, faster-whisper, Next.js

- Transcribes live audio and video streams with speaker attribution, built around a crash-safe job scheduler with fenced leases and checksummed audio windows.
- Delivers events over WebSockets with a transactional outbox and bounded replay, so reconnecting clients catch up without gaps.
- Blocks server-side request forgery with a loopback proxy that rejects non-public addresses on every request and redirect; 90 backend tests and Playwright smoke tests in CI.

SKILLS

Languages: TypeScript, JavaScript, Python, SQL
Frontend: React, Next.js, Vite, Tailwind CSS, GSAP, accessible UI (React Aria, WCAG AA)
Backend and data: Node.js, FastAPI, PostgreSQL, Prisma, Firebase (Firestore, Cloud Functions, Auth), SQLite
Maps and GIS: MapLibre GL, PMTiles vector tiles, GeoJSON, Turf.js, deck.gl
DevOps and ML: Docker, GitHub Actions, Cloudflare Tunnel, Proxmox, Linux · YOLO, PaddleOCR, OpenVINO, Whisper
Practices: security hardening, automated testing (Vitest, Playwright), AI-assisted development (Claude Code, Codex)

EDUCATION

BS Computer Science, Magna Cum Laude · USTP · 2022 – 2026

DOST-SEI JLSS scholar · Alan Turing Award · Founding Vice President, Computer Science Student Society